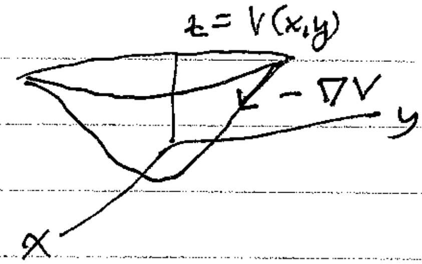
Ruling out limit cycles in $\mathbb{R}^n$: 2 very special cases.

① "Gradient Systems"
② "Systems with a "Lyapunov Function"

① Gradient systems have the form

$$\dot{x} = -\nabla V, \quad x \in \mathbb{R}^n, \quad V = \text{continuously differentiable function } V: \mathbb{R}^n \to \mathbb{R}$$

What is direction of ∇V?



∇V pts. in direction of maximum increase of V.
$F = -\nabla V$ pts in direction of steepest descent.

If $\dot{x} = -\nabla V$, then there are no limit cycles.

If there was a limit cycle $x(t)$, then $x(t) = x(t+T)$ for some (minimal) period $T > 0$.
Compute $V(x(t)) = V(t)$ along the limit cycle, then $\Delta V = V(T) - V(0) = 0$
&

$$\Delta V = \int_0^T \frac{dV}{dt}\, dt = \int_0^T \nabla V \cdot \dot{x}\, dt = \underbrace{-\int_0^T |\nabla V|^2}_{<0}$$

(Can't have $\nabla V = 0$ since that's a fixed pt.)
This is a contradiction $\Rightarrow$ then there is no limit cycle.

Lyapunov Functions

let $\dot{x} = f(x)$ with $f(x^*) = 0$, $x \in \mathbb{R}^n$

If there exists a Lyapunov Function $V(x)$, then x^* is a globally asymptotically stable fixed-pt. (ie. $x(t) \to x^*$ as $t \to \infty$ for all solns) and hence there are no limit cycles.

Lyapunov Function $V(x)$ is defined by the following properties:

1. $V(x) > 0$ for all $x \neq x^*$ and $V(x^*) = 0$

2. $\dot{V} = \frac{d}{dt}(V(x(t))) < 0$ for all $x(t) \neq x^*$,

where $x(t)$ is a soln. to $\dot{x} = f(x)$.

[In practice finding a Lyapunov function is difficult.]

globally asymptotically stable $\lim_{t\to\infty} x(t) = x^*$ for all solns $x(t)$.

Ex. Show that

$$\dot{x} = y - x^3$$
$$\dot{y} = -x - y^3$$

has no closed orbits by constructing a Lyapunov function $V = ax^2 + by^2$ with suitable a, b.

1. Require $a, b > 0 \Rightarrow V \geq 0$ for all $x, y \in \mathbb{R}^2$
& $V = 0$ only at $x^* = (0,0)$

2. $\dot{V} = 2ax\dot{x} + 2by\dot{y}$

$$= 2\left[ax(y - x^3) + by(-x - y^3)\right]$$

$$= 2\left[-ax^4 - by^4 + axy - bxy\right]$$

Choose $a = b = \frac{1}{2}$

$$\dot{V} = -\left[x^4 + y^4\right] < 0 \quad \text{for all} \quad (x,y) \neq x^* = (0,0)$$

Note: we could not use Poincaré index argument to rule out a limit cycle since the origin is a center:

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \qquad \begin{matrix} \tau = 0 \\ \Delta = 1 \end{matrix} \Rightarrow \lambda_{\pm} = \pm i$$

<u>Poincaré-Bendixson Thm.</u> can be used to prove existence of a limit cycle for problems in the phase plane.

Suppose

(1) R is a closed, bounded subset of the plane

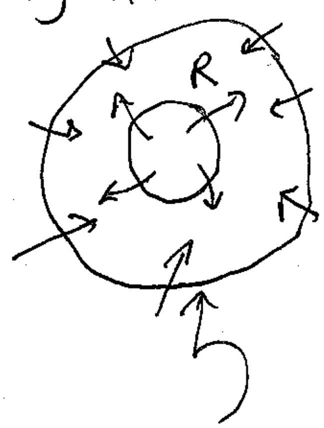
(2) $\dot{x} = f(x)$ is C^1 on open set containing R

(3) R doesn't contain any fixed pts.

(4) There exists a trajectory C confined to R (starts in R, stays in R)

Then, either C is a closed orbit, or it spirals toward a closed orbit as $t \to \infty$. In either case R contains a closed orbit.

Approach used to show (4): use a "trapping region" for R



"trapping region": vector field on boundary pts. into R so trajectories can't escape.

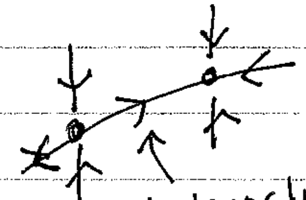
R = annulus, excludes fixed-pt. inside ~~[illegible]~~ limit cycle.

Alternative statement of Poincaré Bendixson:

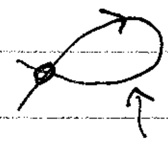
Every non-empty compact ω-limit set of an orbit for a real differentiable dynamical system defined on an open subset of the plane, which has only a finite # of fixed pts. is either

- a fixed pt
- a periodic orbit
- a connected set composed of a finite # of fixed pts. together with homoclinic & heteroclinic orbits connecting them

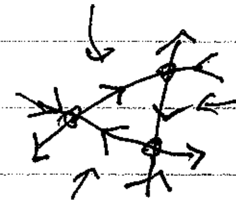
ω-limit sets obtained as $t \to \infty$



heteroclinic orbit



homoclinic orbit



heteroclinic orbits

"There is no chaos in the phase plane"